

FINAL REPORT

Project Name: Sports Hall Booking System By Service Portal

Introduction

a) Project Overview

The Sports Hall Booking System using ServiceNow Service Portal is a web-based application that automates the sports hall booking process. Administrators can efficiently manage approvals and maintain booking records through the ServiceNow platform.

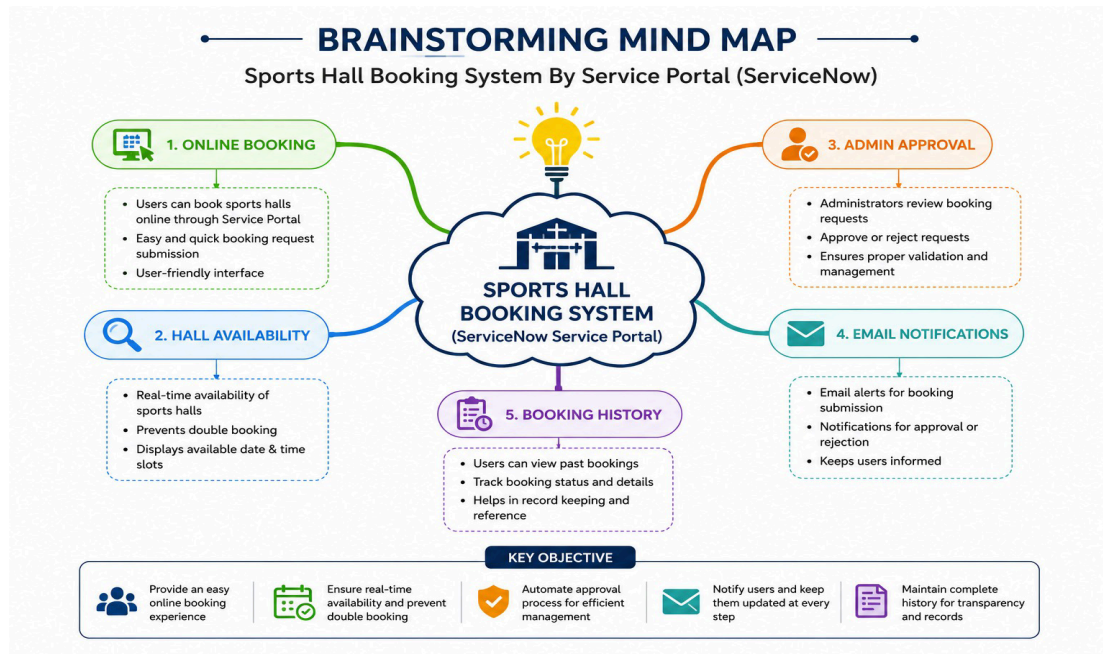
b) Purpose

The purpose of this project is to provide a simple, secure, and efficient online booking system for sports halls.

IDEATION PHASE

1. Brainstorming – Idea Generation & Prioritization

The idea for the Sports Hall Booking System using ServiceNow Service Portal was developed to overcome the problems of manual sports hall booking. The system provides online booking, real-time hall availability, admin approval, booking history, and email notifications.



2. Brainstorming Template

Problem: Manual booking causes delays and scheduling conflicts.

Solution: Develop an online booking system using the ServiceNow Service Portal.

Target Users: Students, Faculty, Sports Department Staff, and Administrator.

Benefits: Easy booking, faster approvals, reduced paperwork, transparent status, and better resource management.

3. Define Problem Statement

The existing manual booking process is slow, inefficient, and prone to double bookings. A centralized ServiceNow-based booking system is needed to automate booking requests and approvals.

4. Problem Statement Template

- **Problem:** Manual sports hall booking.
- **Users:** Students, Faculty, Sports Department.
- **Importance:** Reduces delays and booking conflicts.
- **Solution:** Online booking with automated approvals and notifications.

5. Empathy Map Canvas

User: Students and Faculty.

Says: "I want to book a hall quickly."

Thinks: "Will my booking be approved?"

Does: Checks availability and submits a booking request.

Feels: Frustrated by manual booking but satisfied with quick approvals.

6. Empathy Map Summary

User	Need	Pain Points	Gain
Students & Faculty	Quick online booking	Manual process, delays, double booking	Fast, transparent, and easy booking

Phase 1 Outcome

- Finalized the project idea.
- Identified existing problems.
- Defined the problem statement.
- Analyzed user needs using the Empathy Map.

REQUIREMENTS ANALYSIS

Solution Requirement:

TEAM ID	SWTID-2026-6459
PROJECT NAME	Sports Hall Booking System By Service Portal

Functional Requirements:

Following are the functional Requirements.

FR No	Functional Requirement	Sub Requirements
FR-1	Data Architecture (tables)	Create table (sports hallbooking system),create fields in the table

FR-2	Logics (scheduled job)	Create Scheduled Job(Sports Hall Booking Reminder Alerts) and give the script.
FR-3	Business Rules	Create Business Rule (Calculate Remaining Payment) and give the script
FR-4	Interface Design (Portal)	Create a portal (Portal Branding & Navigation Bar Styling) and give the script
FR-5	Navigation flow	Create a page and select sports and proceed to payment.
FR-6	Data Handling (Events)	Configure email notification for booking confirmation and reminder.

Non-Functional Requirements:

Following are the non functional Requirements

FR NO.	Non Functional Requirements	Description
NRF-1	Business Objectives	Digitize and automate sports hall bookings ServiceNow to eliminate manual coordination, prevent doublebookings, and enable data-driven reporting.
NRF-2	Functional Scope	Configure ServiceNow widgets to handle real-time booking validation, automated manager workflows, and dynamic fee/refund calculations.
NRF-3	Stakeholder Mapping	Stakeholder Mapping stakeholder mapping outlines core booking, approval, and maintenance responsibilities for End Users, Managers, and Facility Administration.
NRF-4	Execution Roadmap	A seven-step project roadmap tracking key milestones from initial requirements gathering and UI design through testing to final deployment.

1. Customer Journey Map

- Login to portal
- Check hall availability
- Submit booking request
- Admin approves/rejects

2. Data Flow Diagram (DFD)

DFD (Level

- User sends booking request
- System stores data
- Admin checks and approves/rejects
- System updates status

3. User Stories

User Story 1

As a Student, I want to log into the Service Portal so that I can book a sports hall.

User Story 2

As a User, I want to view available sports halls so that I can select a suitable date and time.

User Story 3

As a User, I want to submit a booking request so that I can reserve a sports hall online.

User Story 4

As an Administrator, I want to approve or reject booking requests so that I can manage hall availability efficiently.

User Story 5

As a User, I want to receive email notifications so that I know the booking status.

User Story 6

As a User, I want to view my booking history so that I can track previous and upcoming bookings.

4. Solution Requirements

Functional

- Login system
- View halls
- Book halls

Non-Functional

- Secure
- Easy UI
- Reliable

5. Solution Requirements Summary

Input: Login, hall, date, time

Process: Check, validate, approve

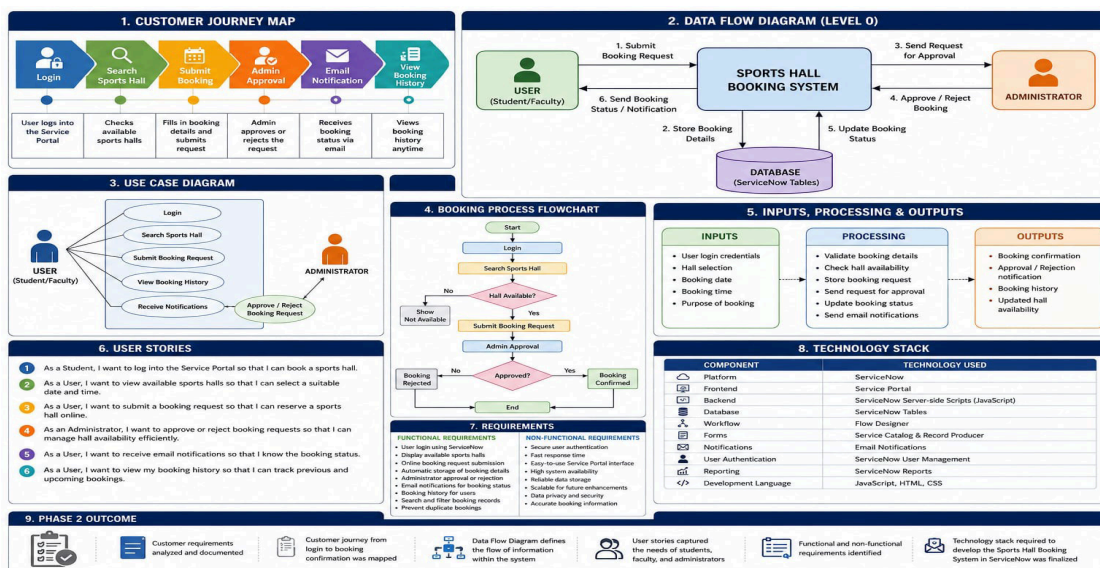
Output: Confirmation, status, history

6. Technology Stack

Component	Technology Used
Platform	ServiceNow
Frontend	Service Portal
Backend	ServiceNow Server-side Scripts
Database	ServiceNow Tables
Workflow	Flow Designer
Forms	Service Catalog & Record Producer
Notifications	Email Notifications
Development Language	JavaScript, HTML, CSS

7. Conclusion

This system makes sports hall booking easy, fast, and automated with proper approval and notifications.



Project Design Phase

1. Problem – Solution Fit Template

Problem

The existing sports hall booking process is manual and inefficient. Users must visit the sports department or contact staff to reserve a sports hall, which often results in delays, booking conflicts, paperwork, and poor record management. **Key**

Challenges

- Manual booking process.
- Double booking of sports halls.

Proposed Solution

The **Sports Hall Booking System** is developed using the **ServiceNow Service Portal** to provide an online platform where students and faculty can book sports halls easily.

Problem–Solution Fit

Problem	Solution
Manual booking	Online booking through Service Portal
Double booking	Availability check before confirmation
Delayed approvals	Automated approval workflow
Paper-based records	Centralized digital database

2. Proposed Solution

Overview

The Sports Hall Booking System is a web-based application built on the ServiceNow Platform. It enables students and faculty to book sports halls through the Service Portal, while administrators manage approvals and maintain booking records.

System Features

User Modul

- Login through ServiceNow.
- View available sports halls.
- Submit booking requests.
- Check booking status.

Administrator Module

- View booking requests.
- Approve or reject bookings.
- Manage sports hall information.

Advantages

- Easy online booking process.
- Faster approval of requests.
- Reduced manual work.

3. Solution Architecture

Architecture Overview

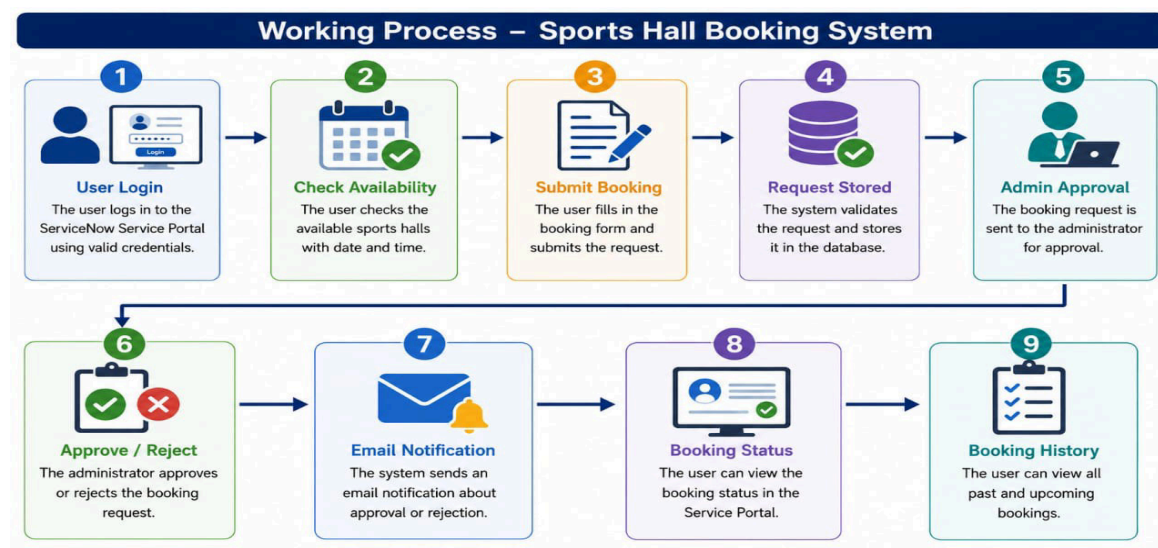
The Sports Hall Booking System follows a simple three-layer architecture.

- **Presentation Layer**
- **Application Layer**
- **Data Layer**

Components Use

Component	Purpose
Service Portal	User interface for booking
ServiceNow Tables	Store booking and hall data
Business Rules	Validate booking requests
Flow Designer	Automate approval workflow
Client Scripts	Improve form interaction
Server-side Scripts	Process booking requests
Email Notifications	Inform users about booking status

Working Process



Phase 3 Outcomes

At the end of the **Project Design Phase**:

- Problems and solutions identified.
- System design completed.
- Three-layer architecture defined.

Project PlanningPhase

1. Planning Logic

Objective:

Plan the project activities, schedule, and resources.

Planning Activities:

Requirement analysis, design, development, testing, deployment, and documentation.

Planning Goals:

Deliver a secure, reliable, and user-friendly Sports Hall Booking System on time. Maintain high system reliability and performance.

2. Project Planning Template

Project Information

Project Title: Sports Hall Booking System using ServiceNow Service Portal

Project Objective: To develop an online sports hall booking application that allows students and faculty to reserve sports halls efficiently while enabling administrators to manage booking approvals and records.

Project Duration: 8 Weeks

Development Platform: ServiceNow

Project Schedule

Phase	Activity	Duration
Phase 1	Ideation and Requirement Analysis	1 Week
Phase 2	System Design	1 Week
Phase 3	Database and Table Creation	1 Week
Phase 4	Service Portal Development	2 Week
Phase 5	Workflow and Approval Configuration	1 Week

Phase 6	Testing and Bug Fixing	1 Week
Phase 7	Deployment and Documentation	1 Week

Resources Required

Software

- ServiceNow Platform
- ServiceNow Service Portal
- Web Browser

Hardware

- Laptop/Desktop
- Internet Connection

3. Project Planning Template (Task Breakdown)

Work Breakdown Structure (WBS)

Task No.	Task	Status
1.	Requirement Analysis	Completed
2.	Project Design	Completed
3.	Create ServiceNow Tables	Completed
4.	Develop Service Portal	Completed
5.	Create Booking Form	Completed
6.	Configure Approval Workflow	Completed
7.	Configure Email Notifications	Completed
8.	Perform System Testing	Completed
9.	Deploy Application	Completed
10.	Prepare Documentation	Completed

Project planning & Scheduling:

Assigned Task to the Group members as shown in below.

Functional Requirements	No.of Activity	Team Members
Business Objectives , Functional Scope	2	Tulugu Urmila
Stack holder Mapping, Execution RoadMap	2	Timmana Varshini
Data Architecture ,Logics	2	Tiddimi Sirisha
Business Rules, Interface Design	2	Vajjula Velangini
Navigation Flow, Data Handling	2	Teppala Pavani

Phase 4 Outcome

A complete project plan with tasks, timeline, resources, risks, and milestones was prepared to ensure the successful development and deployment of the Sports Hall Booking System.



Project Development Phase

Objective

1. Performance Testing

Performance Testing is conducted to evaluate the speed, stability, and reliability of the Sports Hall Booking System developed using the ServiceNow Service Portal.

Service Portal (Sports Zone)

The screenshot shows the ServiceNow Service Portal configuration interface for the 'Sports Zone'. The page includes a navigation bar with 'Service Portal - Sports Zone' and a search bar. A message at the top states: 'Please add the field 'Enable certificate based authentication' using Form Layout to enable the feature. More Details'. The configuration fields are organized into two columns. The left column contains: Title (Sports Zone), URL suffix (sz), Homepage (sports_booking), KB home page, Login page, Logo (Click to add...), Logo Alt Text (Sports Zone), Icon (Click to add...), Quick start config, and CSS variables. The right column contains: Application (Global), 404 page, Catalog home page, Catalog category home page, Main menu, Theme (La Jolla), and Inactive (checkbox). The bottom of the page shows a Windows taskbar with the time 21:18 and date 30-06-2026.

Explanation:

The Sports Zone Service Portal serves as the main interface for students and faculty. Users can log in, access the booking services, and submit sports hall booking requests through a simple and user-friendly portal.

Service Portal (Sports Zone)

This screenshot shows the 'CSS variables' section of the ServiceNow Service Portal configuration for the Sports Zone. The 'CSS variables' field is expanded, showing a code editor with the following CSS:

```
.navbar-brand {
  font-size: 30px !important;
  font-weight: 700 !important;
  letter-spacing: -1.5px !important;
  color: #ffffff !important;
}
.navbar-brand .zone-text {
  color: #ffffff !important;
}
```

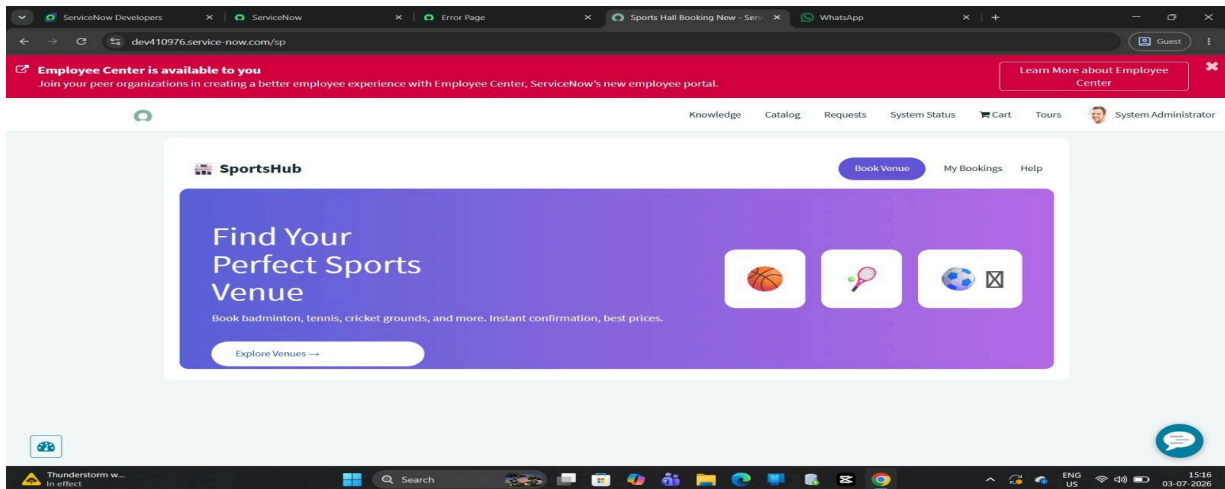
 Below the code editor, there are checkboxes for 'Hide portal name' and 'Support right-to-left languages'. The bottom of the page shows a Windows taskbar with the time 21:30 and date 30-06-2026.

Performance Testing Activities

- Test user login

- Check portal and booking form loading
- Verify booking submission

Navigation Flow



Explanation:

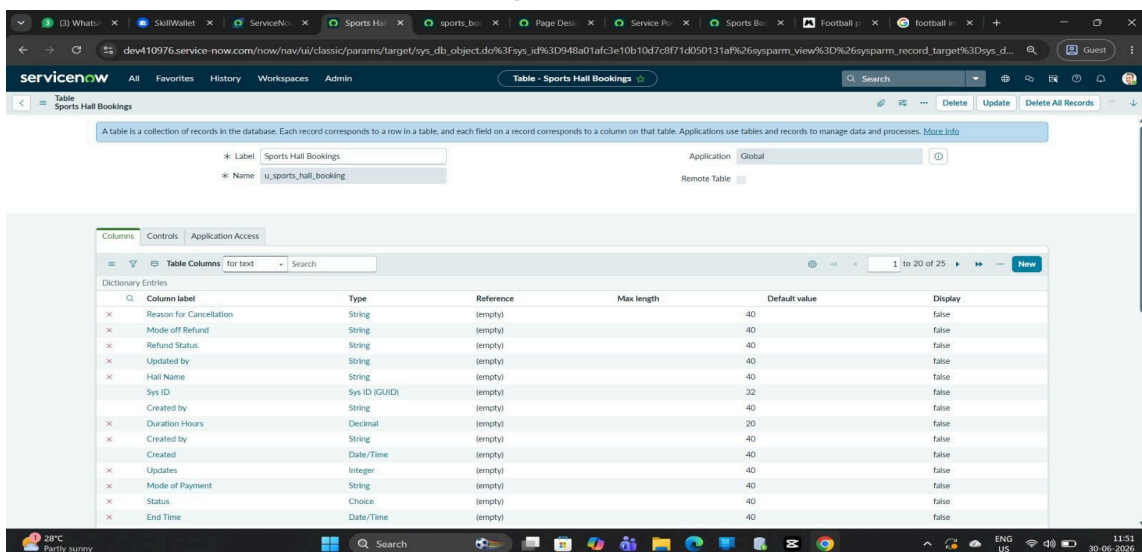
The navigation flow was tested to ensure users can easily move between the Home page, Hall Booking page, Booking History, and other portal pages without errors.

Hall Bookings

Table Explanation:

A custom Hall Bookings table was created in ServiceNow to store all sports hall booking information, including user details, hall information, booking status, and approval records.

Creation of Tables (Hall Bookings)



Creation of Fields (Add after Hall Bookings Table)

The screenshot shows the 'Table Columns' configuration page for the 'Sports Hall Bookings' table in ServiceNow. The table lists various columns with their types, references, max lengths, default values, and display settings. The columns include: Reason for Cancellation (String, 40, false), Mode of Refund (String, 40, false), Refund Status (String, 40, false), Updated by (String, 40, false), Hall Name (String, 40, false), Sys ID (Sys ID (GUID), 32, false), Created by (String, 40, false), Duration Hours (Decimal, 20, false), Created by (String, 40, false), Created (Date/Time, 40, false), Updates (Integer, 40, false), Mode of Payment (String, 40, false), Status (Choice, 40, false), End Time (Date/Time, 40, false), Payment (Currency, 20, false), Purpose (String, 40, false), Start Time (Date/Time, 40, false), Updates (Integer, 40, false), Sys ID (String, 40, false), and Updated by (String, 40, false). The 'Display' column indicates whether each column is visible in the table view.

Column label	Type	Reference	Max length	Default value	Display
Reason for Cancellation	String	(empty)	40	(empty)	false
Mode of Refund	String	(empty)	40	(empty)	false
Refund Status	String	(empty)	40	(empty)	false
Updated by	String	(empty)	40	(empty)	false
Hall Name	String	(empty)	40	(empty)	false
Sys ID	Sys ID (GUID)	(empty)	32	(empty)	false
Created by	String	(empty)	40	(empty)	false
Duration Hours	Decimal	(empty)	20	(empty)	false
Created by	String	(empty)	40	(empty)	false
Created	Date/Time	(empty)	40	(empty)	false
Updates	Integer	(empty)	40	(empty)	false
Mode of Payment	String	(empty)	40	(empty)	false
Status	Choice	(empty)	40	(empty)	false
End Time	Date/Time	(empty)	40	(empty)	false
Payment	Currency	(empty)	20	(empty)	false
Purpose	String	(empty)	40	(empty)	false
Start Time	Date/Time	(empty)	40	(empty)	false
Updates	Integer	(empty)	40	(empty)	false
Sys ID	String	(empty)	40	(empty)	false
Updated by	String	(empty)	40	(empty)	false

Explanation:

Additional fields such as Hall Name, Duration Hours, Transaction Number, Refund Payment, Reminder, and Instructions were created to capture complete booking details and improve booking management.

Test Cases

Test Case	Expected Result	Status
User Login	User logs in successfully	Passed
Load Service Portal	Portal opens successfully	Passed
View Available Sports Halls	Halls displayed correctly	Passed
Submit Booking Request	Booking created successfully	Passed
Approval Workflow	Request sent to administrator	Passed
Email Notification	Notification sent	Passed
Booking History	Records displayed correctly	Passed

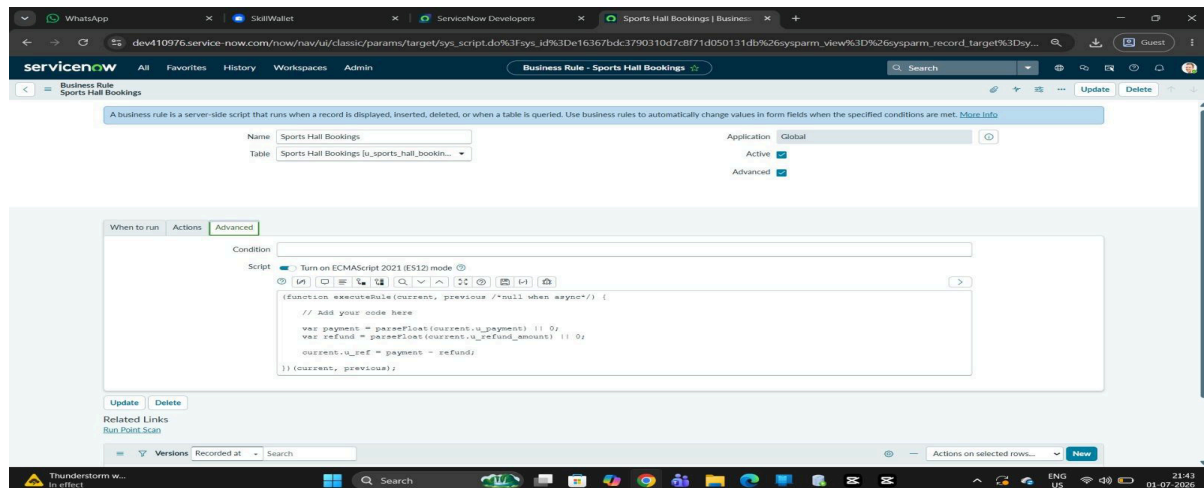
Business Rules

The screenshot shows the 'Business Rule' configuration page for the 'Sports Hall Bookings' table. The rule is named 'Sports Hall Bookings' and is set to run on 'Insert' or 'Update' events. The 'When' condition is set to 'before' with an order of 100. The 'Filter Conditions' section is empty. The 'Actions' section is set to 'Insert', 'Update', 'Delete', and 'Query'. The 'Advanced' tab is selected, showing the 'When to run' and 'Filter Conditions' sections. The 'Update' and 'Delete' buttons are visible at the bottom of the configuration area.

Explanation:

Business Rules were implemented to validate booking requests, prevent duplicate bookings, and automate backend processes whenever a booking record is created or updated.

Business Rules



Scheduled Job

The screenshot shows the ServiceNow Scheduled Jobs list. The table has columns for Name, Active, Class, and Updated. The jobs are listed in descending order of the updated date.

Name	Active	Class	Updated
Search	Search	Search	Search
test import from path ziped	false	Scheduled Data Import	2007-06-11 15:23:40
test import from path ziped tab	false	Scheduled Data Import	2007-06-11 15:23:40
test import from JDBC	false	Scheduled Data Import	2007-06-11 15:23:40
test import from attached	false	Scheduled Data Import	2007-06-11 15:23:40
test import from scp	false	Scheduled Data Import	2007-06-11 15:23:40
test import from path CSV	false	Scheduled Data Import	2007-07-06 14:26:40
test import from path XLS	false	Scheduled Data Import	2007-07-06 14:27:00
Example LDAP Group Import	false	Scheduled Data Import	2008-05-22 13:50:07
Example LDAP User Import	false	Scheduled Data Import	2008-05-22 13:58:36
Synchronous Import Set Completer	true	Scheduled Script Execution	2009-01-03 23:39:56
Server Minimum Statistics	true	Audit	2013-04-08 11:21:03
Unik Server Baseline	true	Audit	2013-04-08 11:21:03
Windows Computer Minimum Statistics	true	Audit	2013-04-08 11:21:24
Data Center Zone Usage	true	Audit	2013-07-29 11:21:42
Outage Data Update	true	Scheduled Script Execution	2013-08-23 16:33:38
Server services statistics	true	Audit	2015-08-24 16:30:36
CMDB Health Dashboard - Correctness Score Calculation	false	Scheduled Script Execution	2015-11-30 12:23:45

Explanation:

A Scheduled Job named Sports Hall – 10 Min Reminder was configured to automatically send reminder notifications to users before their booked sports hall session.

Scheduled Jobs (Sports Hall – 10 Min Reminder)

Scheduled Script Execution - Sports Hall - 10 Min Reminder

Name: Sports Hall - 10 Min Reminder

Active: ☒

Application: Global

Conditional: ☐

For scheduled job types that require an entered time, you have the option to enter an associated time zone. If no time zone is selected, the job will run at the entered time in time zone of the user who entered the time. If 'Use System Time Zone' is selected, the entered time will run in the time zone of the instance running the job.

Run: Periodically

Time zone: Use System Time Zone

Repeat Interval: Days 0 Hours 00 Minutes 30

Advanced: ☒

Starting:

Ending:

Run this script: ☒ Turn on ECMAScript 2021 (ES12) mode

```

1 (function() {
2
3     // Get current datetime
4     var now = new GlideDateTime();
5
6     // Calculate 5 minutes from now
7     var fiveMinutesLater = new GlideDateTime();
8     fiveMinutesLater.add(5 * 60 * 1000); // 5 minutes in milliseconds
9
10    // Calculate 7 minutes from now (window buffer)
11    var sevenMinutesLater = new GlideDateTime();
12    sevenMinutesLater.add(7 * 60 * 1000);
13
14    // Query bookings
15    var gr = new GlideRecord('u_sports_hall_bookings');
16    gr.addQuery('u_status', 'pending');
17    gr.addQuery('u_reminder_sent', false);
18    gr.addQuery('u_start_time', '>=', fiveMinutesLater.getValue());
19    gr.addQuery('u_start_time', '<', sevenMinutesLater.getValue());
20    gr.query();
21    while (gr.next()) {
22
23

```

Configure Events

Scheduled Script Execution - Sports Hall - 10 Min Reminder

Run this script: ☒ Turn on ECMAScript 2021 (ES12) mode

```

1 (function() {
2
3     // Get current datetime
4     var now = new GlideDateTime();
5
6     // Calculate 5 minutes from now
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10    // Calculate 7 minutes from now (window buffer)
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12    sevenMinutesLater.add(7 * 60 * 1000);
13
14    // Query bookings
15    var gr = new GlideRecord('u_sports_hall_bookings');
16    gr.addQuery('u_status', 'pending');
17    gr.addQuery('u_reminder_sent', false);
18    gr.addQuery('u_start_time', '>=', fiveMinutesLater.getValue());
19    gr.addQuery('u_start_time', '<', sevenMinutesLater.getValue());
20    gr.query();
21    while (gr.next()) {
22
23

```

Execute Now Update Delete

Related Links
[View Dashboard](#)
[Run Point Scan](#)

Explanation:

The event sports.hall.reminder was configured to trigger reminder notifications whenever the scheduled job executes.

Configure Events (sports.hall.reminder)

Notification - Booking confirmation reminder

Name: Booking confirmation reminder

Type: EMAIL

Table: Sports Hall Bookings [u_sports_hall_bookings]

Category: Uncategorized

Description:

A digest combines and summarises multiple updates for a single or multiple target records within a set time interval.

Allow Digest: ☐

Who will receive:

What it will contain:

Send when:

When there are multiple notifications in the Outbox that are triggered from the same Table and Notification and addressed to the same recipients, notifications with the higher weight are sent first. (0 being the default and highest priority of weight)

Use an Advanced condition to send a notification based on the current email record, changing field values, or system properties. To send the notification, your advanced condition script must set a global answer variable to true.

The Advanced condition script has access to the following global variables:

- current - contains the current record from the table to which the notification is linked
- event - contains the event that triggered the notification

Send when: Event is fired

Event name: sports.hall.reminder

Conditions: Add Filter Condition Add OR Clause

Choose field: Operator: Value:

Advanced condition: ☒ Turn on ECMAScript 2021 (ES12) mode

Weight:

Explanation:

Email Notifications were configured to automatically send booking confirmations and reminder messages to users after booking submission and approval.

Performance Test Results

Parameter	Result
Login Performance	Successful
Booking Request Processing	Successful
Workflow Execution	Successful
Email Notification	Delivered Successfully
Database Response	Accurate

Performance Testing Summary

The Sports Hall Booking System performed successfully during testing. All major functionalities, including booking creation, approval workflow, reminder notifications, and email delivery, worked as expected.

2. User Acceptance Testing (UAT)

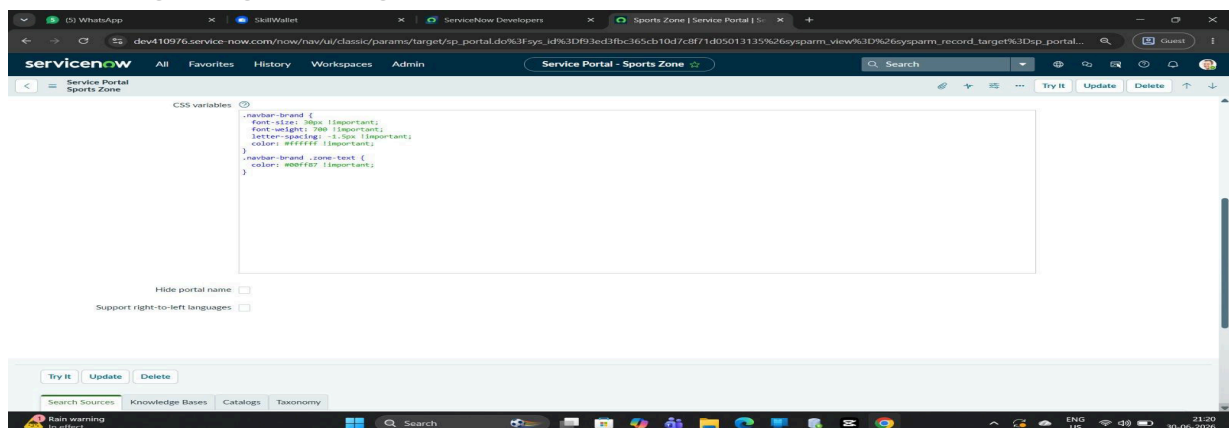
Objective

Ensure that the Sports Hall Booking System meets user requirements before deployment.

Users Involved

- Faculty Members
- Sports Administrator
- System Administrator

Home Page Logo Configuration



Explanation:

The Home Page Logo Configuration provides the final branded interface of the Sports Zone portal, offering users a professional and easy-to-use environment during User Acceptance Testing.

User Acceptance Test Scenarios

- User login
- View hall availability
- Admin approval
- Email notification

UAT Feedback**Positive Feedback**

- User-friendly interface
- Quick booking process
- Useful email notifications

User Acceptance Test Results

Evaluation Criteria	Result
Functional Requirements	Met
Ease of Use	Excellent
System Reliability	High
Booking Accuracy	Accurate
Workflow Functionality	Successful
User Satisfaction	High

Phase 5 Outcome

Performance testing was completed successfully. Core features, including booking, approvals, reminders, and notifications, were validated.

Project Documentation Phase

1. Final Report Template

Project Title

Sports Hall Booking System using ServiceNow Service Portal

Project Abstract

The Sports Hall Booking System is a web-based application developed using the ServiceNow Service Portal. It automates the process of booking sports halls for students and faculty.

Introduction

The purpose of this project is to develop a centralized system for managing sports hall bookings in an educational institution.

Objectives

- To automates sports hall booking process.
- To eliminate manual errors and double booking.
- To provide real-time availability of halls.
- To improve communication through notifications.

System Overview

The system is built using ServiceNow Service Portal, which provides a user-friendly interface for booking.

Modules

- User Module
- Administrator Module
- Booking Module

Conclusion

The system successfully automates sports hall booking and improves efficiency, transparency, and user satisfaction.

2. FSD (Functional Specification Document) Format

Introduction

This document defines the functional requirements of the Sports Hall Booking System developed using ServiceNow.

Functional Requirements

- User Functions
- Admin Functions

Security

- Role-based access control
- Secure login authentication
- Data protection in ServiceNow tables

3. FSD Documentation Format System Architecture

Layers

- Presentation Layer (Service Portal)
- Application Layer (Business Rules, Flow Designer)
- Data Layer (ServiceNow Tables)

Data Design

Tables Used

- User Table
- Sports Hall Table
- Booking Table
- Approval Table

Workflow Design

Trigger: New booking request

Condition: Hall availability check

Action

- Send to approval
- Update status
- Send email notification

Business Rules

- Prevent duplicate bookings
- Validate required fields
- Ensure correct date/time selection

Phase 6 Outcome

A complete final project report was prepared with objectives, modules, and conclusions..